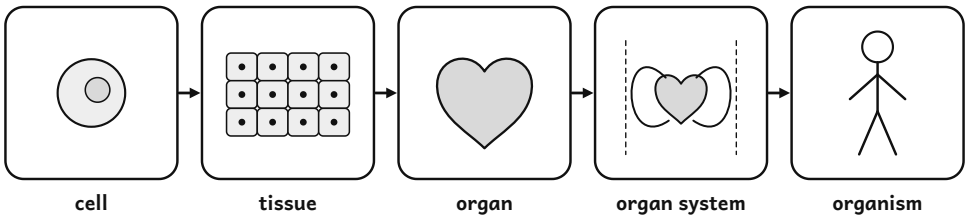
LIVING THINGS

Cells, Tissues and Organs

Cells as building blocks, specialised cells and the levels of organisation in the human body



In this pack

📖 **Learn it:** Cells: the building blocks of life ★☆☆
Fact check: Cells and specialised cells ★☆☆
Think it through: Shape and job ★★☆
Investigate: Looking at cells with a microscope ★★☆
Real-world case study: Giving blood: cells that save lives ★★★
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This pack belongs to: _____ **Class:** _____

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LEARN IT · A2 · CELLS

Cells: the building blocks of life

Name: Class: Date:

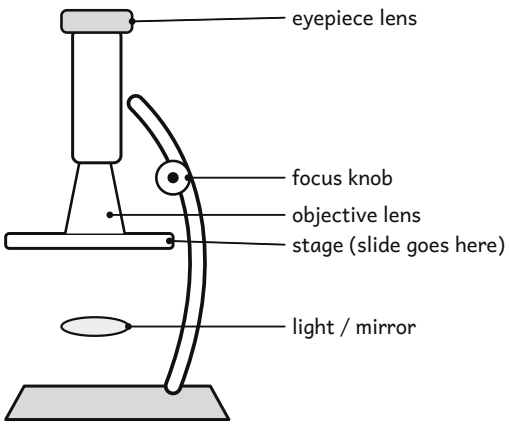
Learning goals

- Recall that all living things are made of cells.
- Describe specialised cells in the human body and their jobs.
- Explain the levels of organisation: cell, tissue, organ, organ system, organism.

A house is built from thousands of bricks. Your body is built from about **30 trillion** tiny units called **cells**. Every living thing, from a mango tree to a tiger, is made of cells. Most cells are far too small to see without a **microscope**.

Who discovered cells?

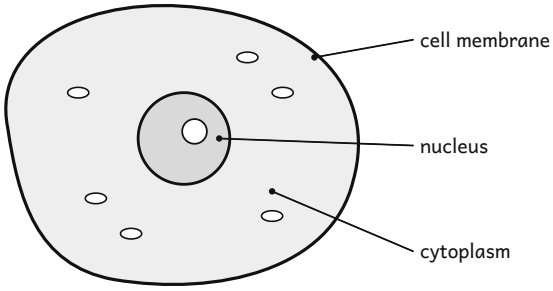
In 1665 the English scientist **Robert Hooke** looked at a thin slice of cork through a microscope he had built. He saw rows of tiny boxes that reminded him of small rooms, so he called them **cells**. A few years later, Antonie van Leeuwenhoek in the Netherlands made better lenses and saw living cells, such as blood cells and tiny creatures swimming in pond water.



A school microscope. The lenses make things look much bigger.

What is inside a cell?

Animal cells have three main parts. The **cell membrane** is a thin skin that holds the cell together and controls what goes in and out. The **cytoplasm** is a jelly where the cell's work happens. The **nucleus** is the control centre: it holds the instructions that tell the cell what to do. Plant cells also have a tough **cell wall** and green **chloroplasts** that trap light.

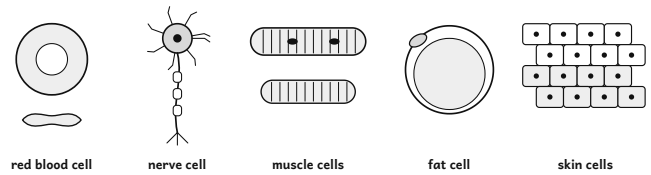


An animal cell.

Specialised cells

Cells come in different shapes because they do different jobs. Their shape helps them do their job well.

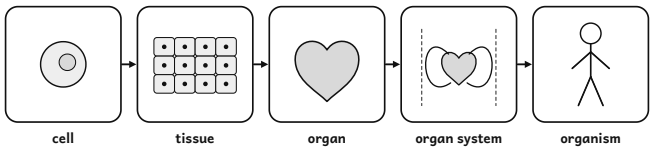
- **Red blood cells** are round and dented in the middle, like a doughnut without a hole. This gives them a large surface to carry oxygen, and they can squeeze through tiny blood vessels.
- **Nerve cells** are long and thin, with branches, to carry messages quickly over long distances.
- **Muscle cells** are long and can get shorter (contract) to move the body.
- **Fat cells** store energy and help keep us warm.
- **Skin cells** fit tightly together like tiles to make a protective layer.



Some specialised cells in the human body (not drawn to scale).

From cells to a whole body

Cells of the same kind group together to make a **tissue**, such as muscle tissue. Different tissues work together to make an **organ**, such as the heart. Organs work together in an **organ system**, such as the circulatory system. All the organ systems together make an **organism**: a living thing, like you!



Levels of organisation in the human body.

Think like a scientist

Think of a school: one **learner** (cell) belongs to a **class** (tissue), classes make a **year group** (organ), year groups make the **school** (organ system)... What would the organism be?

Did you know?

Your body makes about 2 million new red blood cells **every second**. Each one lives for about 120 days.

Key words

cell	the smallest building block of a living thing
nucleus	control centre of a cell
cell membrane	thin layer around a cell that controls what goes in and out
cytoplasm	jelly inside a cell where its work happens
specialised cell	a cell with a special shape for a special job
tissue	a group of similar cells doing the same job
organ	different tissues working together
organism	a whole living thing
microscope	instrument that makes small things look bigger

FACT CHECK · A2 · CELLS

Cells and specialised cells

★☆☆
Grade 5

Name: Class: Date:

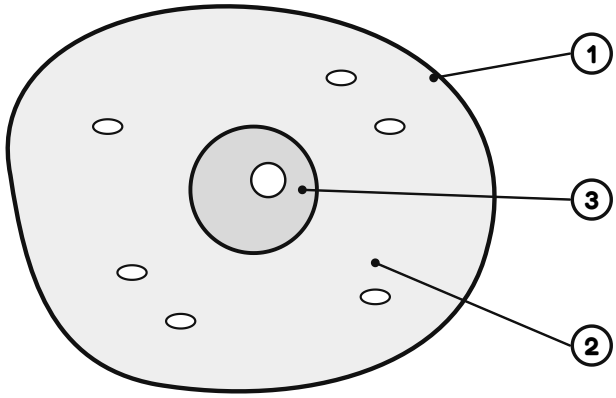
Learning goals

- Label a cell.
- Match cells to their jobs.
- Order the levels of organisation.

1 **Label the parts of the animal cell.** [3]

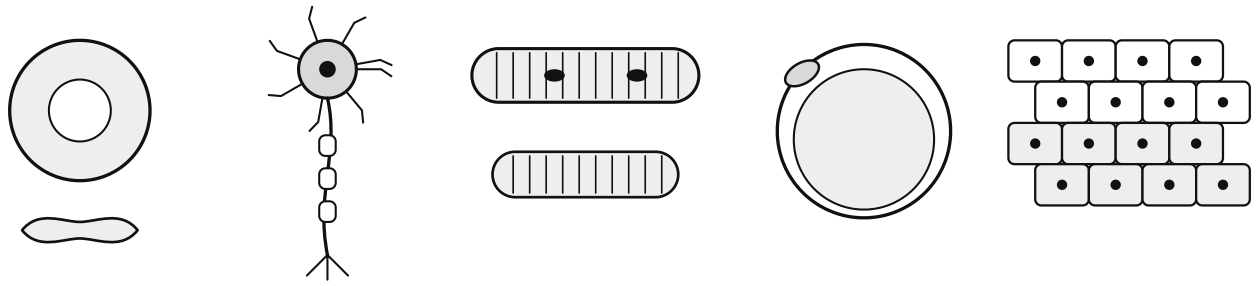
LABELS

cytoplasm **nucleus** **cell membrane**



1 _____ 2 _____ 3 _____

2 **Write the letter of each cell next to its name.** [10]



A B C D E

Cell	Letter	Its job
red blood cell		
nerve cell		
muscle cell		
fat cell		
skin cell		

3 **Number the levels of organisation from smallest (1) to largest (5).** [2]

- ☐ cell
- ☐ organ
- ☐ organ system
- ☐ organism
- ☐ tissue

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. All living things are made of cells.			
2. The nucleus controls what the cell does.			
3. A tissue is made of many different organs.			
4. Robert Hooke named cells after looking at cork.			

THINK IT THROUGH · A2 · CELLS

Shape and job

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain how a cell's shape suits its job.
- Use the levels of organisation to explain how the body is built.

1 **Explain how the shape of each cell helps it do its job.** [6]

a) nerve cell

b) red blood cell

c) skin cells

2 **Complete the table for the digestive system.** [5]

Level	Example
cell	
tissue	
organ	
organ system	
organism	

3 **Rumana says, "If one organ stops working, the whole organism is in danger." Use the idea of organ systems to explain whether she is right.** [2]

4 **A plant cell has a cell wall and chloroplasts. An animal cell does not. Suggest why plant cells need these.** [2]

INVESTIGATE · A2 · CELLS

Looking at cells with a microscope

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Use a microscope safely.
- Calculate magnification.
- Observe, draw and compare cells.

You will need

microscope, a thin piece of onion skin, a glass slide, a cover slip, a drop of iodine (stain), a pencil

Stay safe

Handle glass slides carefully. Iodine stains skin and clothes. Never point a microscope mirror at the Sun.

1 **Magnification tells us how many times bigger something looks. Multiply the eyepiece by the objective lens.** [3]

Eyepiece lens	Objective lens	Total magnification
×10	×4	
×10	×10	
×10	×40	

2 **Follow the steps to look at onion cells. Then draw what you see.** [5]

1. Peel a thin, see-through layer from the inside of an onion piece.
2. Lay it flat on the slide and add one drop of iodine.
3. Lower a cover slip gently on top.
4. Start with the smallest objective lens, focus, then try a bigger one.

My drawing of onion cells (×100)

a) Describe the shape of the cells.

.....

.....

b) Which part of the cell did the iodine make easier to see?

.....

.....

3 **Compare onion cells with cheek cells (animal cells). Fill in the table.** [3]

Feature	Onion cell (plant)	Cheek cell (animal)
shape		
cell wall?		
nucleus?		

4 **Why do scientists add a stain such as iodine?** [1]

.....

.....

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A2 · CELLS

Giving blood: cells that save lives

★★★
Grade 5

Name: **Class:** **Date:**

Learning goals

- Explain the jobs of the parts of blood.
- Use data and reasoning about a real health issue.
- Discuss why people choose to help others.

Science · Health · Maths · Ethics · Community

Blood looks like a red liquid, but it is really a mixture of cells floating in a yellow liquid called **plasma**. **Red blood cells** carry oxygen. **White blood cells** fight germs. Tiny **platelets** help blood clot to stop bleeding.

In Bangladesh, hospitals need blood every day for accident victims, mothers giving birth and people having operations. Many children have **thalassaemia**, an inherited condition in which the body cannot make enough healthy red blood cells. Many of these children need a blood transfusion every few weeks.

Healthy adults can give (donate) blood safely. In 1977, students at Dhaka Medical College started **Sandhani**, a volunteer group that encourages young people to give blood. Today there are many blood donor groups across the country. The body replaces the donated blood: the liquid within a day or two, and the red blood cells within a few weeks.

Information from general medical sources; donors must meet age and health rules set by their blood bank.

1 **Match each part of blood to its job.** [4]

Draw a line to match each pair, or write the letter.

red blood cells ●	● **A** carry oxygen
white blood cells ●	● **B** help blood clot
platelets ●	● **C** fight germs
plasma ●	● **D** liquid that carries the cells and nutrients

2 **A person usually gives about 450 ml of blood. An adult has about 5 litres (5000 ml) of blood.** [2]

a) How much blood is left in the donor's body straight after giving?

.....

b) About what fraction of their blood did they give? Circle: one half · one tenth · one hundredth

.....

3 **Why might a child with thalassaemia feel tired and weak? Use the word *oxygen*.** [2]

.....

.....

4 **Only healthy adults can give blood. Suggest two reasons for this rule.** [2]

5 **Why do you think thousands of students volunteer to give blood to people they will never meet? Give your opinion.** [2]

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

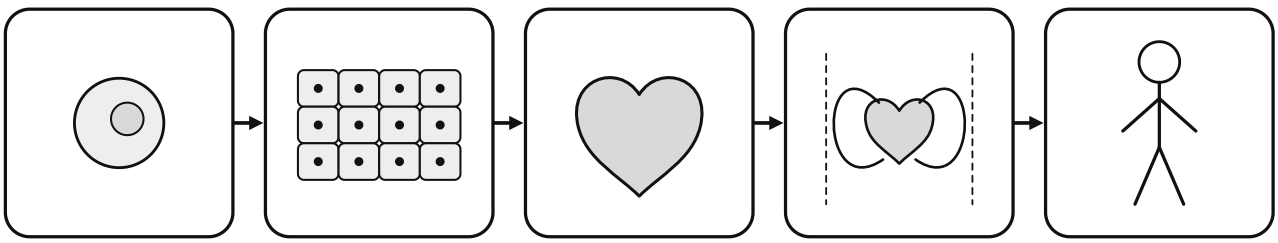
R	A	E	J	T	I	S	S	U	E	D
N	J	I	T	U	Y	C	P	R	L	F
B	A	K	K	E	P	S	S	R	P	A
H	W	G	J	O	R	G	A	N	B	T
M	S	Y	S	T	E	M	R	Y	N	F
E	I	N	W	G	I	N	T	H	E	M
L	S	K	I	N	H	T	J	N	R	U
G	V	B	L	O	O	D	L	N	V	S
B	N	U	C	L	E	U	S	K	E	C
M	O	C	C	B	U	A	V	J	G	L
O	P	C	E	L	L	S	M	K	B	E

blood **cell** **fat** **muscle** **nerve** **nucleus** **organ** **skin** **system** **tissue**

2 Label the levels of organisation. [5]

LABELS

cell **tissue** **organ system** **organ** **organism**



1 _____ 2 _____ 3 _____

4 _____ 5 _____

3 Which cell carries messages around the body? [1]

- ◯ red blood cell
- ◯ nerve cell
- ◯ fat cell
- ◯ skin cell

4 **The heart is made of muscle tissue, nerve tissue and other tissues. This makes the heart:** [1]

- ◯ a cell
- ◯ a tissue
- ◯ an organ
- ◯ an organism

5 **Challenge. Why can we not see cells with our eyes alone, but we can see an organ like the heart?** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain that all living things are made of cells	◯	◯	◯
label the parts of an animal cell	◯	◯	◯
describe five specialised cells and their jobs	◯	◯	◯
order cell, tissue, organ, organ system, organism	◯	◯	◯
use a microscope and calculate magnification	◯	◯	◯
explain the jobs of the parts of blood	◯	◯	◯